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Document	Project Assignment	Project Management 412 Assignment 3
Date	2026-04-24	
Revision	3.1	

Project Assignment Number: 3

1 DESCRIPTION OF ASSIGNMENT

Report on your project's progress, by performing an Earned Value Analysis for the project up to the end of period 7. Organise a Project Progress ("Review") meeting with your team and submit a Project Progress Report evaluating both the Simulated Project and your team's real-life project execution and multi-disciplinary group work. The Simulated Project performance assessment should include an Earned Value Analysis for the project.

1.1 Earned Value Analysis Spreadsheet

Develop an Excel Spreadsheet, detailing the progress of the simulated project up to the end of Period 7, for each period, making use of Earned Value Analysis (EVA) for your simulated project. The following aspects need to be included in the EVA:

- **Use as analysis dates, a date at the close of business (COB), which is five (5) working days before the actual completion date of the simulated period, for each period.**
- Each period must be on a separate sheet in the Excel file
- On a separate summary sheet in the Excel file (make it the first sheet) create a progress dashboard for any selectable period. As a minimum:
 - Show the calculation of AC, PV and EV for the end of the **selected Period** (end of period min five (5) work days, COB)
 - Show the calculation of SPI and CPI for the end of the **selected Period** (end of period min five (5) work days, COB)
 - Show the calculation of likely project cost at completion, based on data up to the end of the **selected Period** (end of period min five (5) work days, COB)
 - Plot AC and EV for all periods up to the selected Period
 - Plot PV and EV for all periods up to the selected Period
 - Plot SPI vs CPI for all periods up to the selected Period
 - Include on this summary sheet/dashboard key graphs and information reflecting the status of your project as at the end of the selected Period (min 5 working days before actual completion)

Bonus points will be earned if you include Training and Management activities in your Earned Value Analysis.

It should be noted that the EVA spreadsheet and supporting calculations will be assessed in detail. In addition, the data included in the dashboard sheet (as at the end of the selected period min 5 working days) will be assessed against your updated *MS Project* file. Lastly, the EVA up to period 7 will be assessed against the project progress report and updated *MS Project* file.

1.2 Project Review Meeting.

Organise and conduct a Project Review meeting with your team. During this meeting, you must evaluate both the Simulated Project, as well as your team's real-life Group Project Execution that you performed by taking periodic decisions and submitting assignments. Ensure that this review meeting is thoroughly minuted, and include the minutes in the report.

1.3 Project Progress Report

Develop and submit a project progress report that addresses both the **Simulated** project that are being executed via Sim4Projects, as well as the **Real-life Group Project Execution** that you as a multi-disciplinary team are performed by taking periodic decisions and submitting assignments.

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1.3.1 *Simulated Project section of report*

The section of the **progress report** for the **Simulated Project** needs to include and discuss the following:

- Project Performance Review (Use Earned Value management where appropriate)
 - Use your Earned Value Analysis for all the periods (**minus five (5) working days for all periods**), and base your discussions on the EVA results up to period 7 where applicable.
 - Include an 'Actual Cost' vs Budget Analysis and Review, and discuss the major variances using your EVA results as the basis for discussion. Keep in mind that since this is a progress report, you should refer to the planned cashflow to date, rather than the total project budget. What is the Estimated Cost at Completion and projected cost variance and why?
 - Include a Schedule Analysis and Review, and discuss the major variances in milestones (actual vs plan) using your EVA results as the basis for discussion. Include an estimate of how far ahead/behind you are with your schedule, and the implications on the Project Plan.
- Team Development
 - Review and discuss your Resource Strategy to date – what was it, and did it work?
 - Did you develop your resources and what was the effect?
 - Did you adapt your resources strategy during the initial 7 periods of the project? If so, why and what was the outcome?
 - If you had to re-do a similar project, what would you do differently in terms of your resource strategy and why?
- Risk Management Plan (RMP) Success
 - Update the Risk Register submitted in Assignment 2 with unwanted events that occurred during the simulation up to period 7, (if not included in original Risk Register). Provide a copy of the prioritised risk register in the Appendix to the report.
 - Based on the unforeseen events up to period 7, how successful was your Risk Management Strategy?
 - The team needs to demonstrate their ability to successfully manage the project risks as a multi-disciplinary team.

1.3.2 *Real-Life Group Project Execution and Multi-Disciplinary Group Work section of report*

The section of the progress report for the **Real-Life Group Project Execution** needs to include and discuss the following:

- Lessons Learnt:
 - What did you as a group learn from the project?
 - Did it help to apply the techniques, which you learnt in theory, in practice?
 - In hindsight, what would you as a team do differently with respect to managing the simulated project and managing the workload of the real-life team?
 - As a group, reflect on the experience of working together as a multi-disciplinary team, including how you managed team dynamics as a result of different personalities and skills in the team. Reflect on the integration of the multi-disciplinary team and how the various engineering functions worked together to achieve the final product.
- Individual Review of Multi-Disciplinary Group Work
 - Each Team member (under the heading of his/her name) should write at least one full page in the report (preferably in the appendices), explaining:
 - What you learnt about yourself as an individual when tasked to function in a multi-disciplinary group environment. Factors to consider could include

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level of contributions, time management, attitude, preparedness, quality of work, ability to work with others etc.

- What would you do differently the next time you work in a multi-disciplinary team?
- What competencies and skills did you observe from members of other disciplines and/or individuals which could be of value to you in terms of project management?
- How, in the future, would you combine the strengths of the various disciplines and/or individuals to ensure successful project execution?
- **NB: the individual review is a standalone part of the assessment to evaluate a GA8. Therefore, it is critical that each student's 1-page submission is included. Not submitting it, may cause an individual to fail the module, irrespective of the quality of the group's submission.**
- Signed minutes of the Project Progress Review meeting.
- Submit signed minutes of all team meetings not yet submitted. These minutes shall clearly describe decisions taken and allocated actions. Ensure that the multi-disciplinary aspects of your project team are taken into consideration.

2 EXPECTATIONS AND OBJECTIVES

The Project Progress Report shall be outlined clearly and the conclusion and recommendations must be clearly shown and discussed in the report.

The team need to demonstrate their ability to use the progress information, as contained in the project plan, to derive a detailed Earned Value Analysis **from first principles**. This should demonstrate a full understanding of Earned Value Analysis and the complexities that may arise when performing it.

The team also needs to demonstrate an ability to use the Baseline Plan, take progress into account and evaluate this progress over all 7 periods against the budget, while making use of Earned Value Analysis. In particular, the team need to demonstrate their ability to manage their plan successfully, by:

- updating their plan in *MS Project*,
- assigning additional resources,
- continually levelling the resources,
- analysing and refining their plan to determine the impact on subsequent periods, and
- creating a dashboard for the simulated project's progress up to any selected Period in a dashboard style, using the data provided.

By also analysing and discussing progress during the Real-life Group Project Execution, the group demonstrates an ability to critically self-assess their success as a project team.

By submitting minutes of meetings in the appendix of the report, the team demonstrates an ability to organise the planning of their simulated project in a multi multi-disciplinary context.

Please take note of the mandatory ECSA Exit Level outcomes in section 6 of the Study Guide.

3 DELIVERABLES

- A project progress report covering:
 - the Simulated Project,
 - the Real Life Group Project Execution, and
 - the project progress report must be complete and should include everything in a single PDF file, including relevant printouts from *Excel* (such as the risk register) **and** *MS Project*. This document will form part of the final assessment

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documentation that will be provided to the internal and external examiners at the end of the semester.

- An Excel Spreadsheet, submitted electronically, containing the Earned Value Analysis as performed, for all periods up to period 7. All Earned Value calculations must be performed in Excel, to demonstrate your full understanding of the Earned Value calculation mechanism. The Spreadsheet should also include a summary dashboard sheet (as first sheet) showing EVA calculations, graphs and progress elements up to any selected Period (i.e., as if progress feedback was given at the end of selected Period).
- An updated *MS Project* plan, based on your baseline plan (in *MS Project* format), with all actual progress up to the end of Period 7 incorporated, as well as resource assignments for future periods. This must be submitted only electronically via STEMLearn.
- A colleague assessment appraisal (Buddy Rating) to be done online with the login information provided.

4 SUBMISSION REQUIREMENTS

Deadline	<i>8 May 2026:</i> • <i>Electronic Submission¹: Before 23h55 on STEMLearn.</i> <i>8 May 2026:</i> • <i>3rd Buddy Rating Submission: Before 23h55.</i>
Method	<i>1) A Project Review Report for the Simulated Project as well as for the Real-Life Group Project Execution: Electronically submitted as *.pdf document via the STEMLearn Turnitin mechanism.</i> <i>2) Excel Spreadsheet with Earned Value Calculations up to Period 7, including summary/dashboard sheet up to any selected Period: Electronically as an Excel file via STEMLearn.</i> <i>3) Updated MS Project File up to Period 7: Electronically as a MS Project file via STEMLearn.</i> <i>4) Colleague Assessment using the online “Buddy Rating” system.</i>
Point of submission	<i>Electronically on STEMLearn– make sure you use the correct link for each deliverable.</i>
Document requirements	<i>The body of the Report document (excluding title page, table of contents and appendices) should not exceed 8 pages (11pt single line spacing, with open lines between paragraphs).</i> <i>Ensure that the report is of a professional nature – typical of what you would be expected of you as professional engineer in practice. Use a professional technical language; proper use of references; a title page with team name, team ID and team members; page numbers, paragraph numbers, etc.</i> <i>Ensure that each student’s individual review of the multidisciplinary group work appears under their own name as heading (refer to 1.3.2 in this document).</i> <i>If the electronic submissions are not named correctly or in the correct format, then a penalty of 5% per file will be applied. As an example, for group A-1, the files should be submitted as:</i> <i>Group A1 Assignment 3.pdf, Group A1 Assignment 3.xlsx & Group A1 Assignment 3.mpp).</i>

¹ The permanent Group Leader is responsible for submitting the electronic files.

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Late submissions	<i>5% penalty for every hour, or part thereof, after the deadline.</i> <i>No submission will be accepted after 06h55 on the day after submission.</i> <i>The buddy rating penalty as specified in the main assignment document will be applied to the individual project mark for all assignment submissions.</i>
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